

Does An Interactive Diegetic Navigation Aid in Virtual Reality Have an Effect on Enjoyment Levels?

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1. Introduction

Over the years there has been a rapid growth in the number of people playing games, according to a report published by DFC intelligence (Conor, 2021), it shows that as at mid-2020 about 2.1 billion people played video games, while the entire population of the world is 8 billion, which means that about 40% of the entire population is playing video games. Personal computer owners that are also using their devices to play video games are at 48% of the population (Webindex). Video games have been a household tool of entertainment and for educational purposes. It was noted that two-third of American parents attested to the fact that educational games made it easy for their children to learn during the long-distance teaching period, which accosted the number of parents that play video games with their children to be at 75% (Mike Snider, 2021).

Playing video games has increasingly become a way of enjoyment and entertainment, one of the key factors that makes video games interesting to play is the navigation of the players in the scene, there are diverse ways game designers have designed how they want the players to experience the gameplay. Navigation in video games has been implemented in unusual ways depending on the design that suits the game, the techniques include minimap, screenspace maps, waypoint graphics (such as arrows), graphical elements that highlight certain locations.

Virtual reality is on the rise in the experience of video games, with a lot of designers and developers since early 2016 when virtual reality began to attract attention and video games are developed to suit it. According to (Dautovic, 2021) 40.1% of the compound annual growth rate of the global gaming and entertainment market is linked to Virtual reality. In the advent of the development of virtual reality based games, (Shafer et al., 2019) noted that spatial presence is a valuable factor when it comes to the enjoyment of video games. Spatial presence is defined as the feeling that the user is part of the environment's built-in virtual world. One of the factors that can increase spatial presence is the introduction of interactive diegetic navigation aids within the game scene.

Although, there have been several VR (Virtual Reality) games that incorporated two navigation components into their gameplay: the virtual character (either embodied through the first-person perspective or the third-person perspective) and the world or level itself. Usually using certain methods of interaction (keyboard and mouse or controllers) the player controls the movement of the virtual player and traversal decisions of the virtual player (Sousa Santos et. al., 2008; Bowman et. al, 2001). Navigation in a game play is bedrock to which the player experience is anchored, bad navigation can spoil a good game and a good game level can be marred by navigation or simulations.

For a player to enjoy a game and the game to be termed successful, the navigation aids in the game play can communicate right back to the player. Furthermore, in this study, we are exploring the effects of interactive diegetic navigation to enjoyment in virtual reality game play. This paper covers 4 areas which are: literature review (related study), methodology, study design and results.

2. Literature Review in Video Games

2.1. Navigation in Video Games

Navigating virtual world within a video game has always been one of the major cornerstones of game design. The current generation of video games offer a multitude of complex landscapes and environments for a player to explore and with that comes the additional task of providing the player with certain aids that can help them traverse the landscapes and reach their intended destinations for a better gameplay experience. Bowman et al. (2001) in one of the earliest research projects on navigation in virtual environments states two major components: travel, which is the motor skill related component, and wayfinding, which is the cognitive skill related component.

Navigation in video games usually incorporates the following two components: the virtual character (either embodied through the first-person perspective or the third person perspective) and the world or level itself. Using certain methods of interaction (keyboard and mouse or controllers) the player can control the movement of the virtual player and traversal decisions of the virtual player. Usually, there is a certain intent and motivation behind how the players move around in a virtual environment (Sousa Santos et. al., 2008; Bowman et. al, 2001). In that context, the navigation component of the entire game or simulation can make or break the impact that the game or simulation is looking to deliver. Due to this aspect, navigation is one of the first concepts that is addressed when developing levels, and certain techniques are used to support the player's navigational needs. These can range from a wide variety of techniques such as minimaps, waypoint graphics, NPC interactions, or having components built into the level design to help guide the player to take certain decisions for reaching their destinations (Solheim, 2021; Chen et. al, 2004). For the sake of this paper, these techniques have been brought under the umbrella term of “navigation aids” in video games, since they assist the player in navigating the virtual environment and enhance their experience by making it more immersive.



Figure 1: An example of a screenspace map (left) and minimap (right) from the game *Sleeping Dogs* (Square Enix, 2012)

2.2. User Interface Design in Video Games

Navigation aids can be attributed to being any technique implemented in a game or a virtual simulation that can aid the player to traverse the virtual environment in the game level or simulation. A wide variety of techniques have been implemented by developers to achieve this goal of navigation, some of the most common being player interface elements such as a minimap, screenspace maps, waypoint graphics (such as arrows), graphical elements that highlight certain locations or routes to be taken, etc. In traditional video games (i.e., games consumed through flat paneled display devices), all these elements are contained in what has come to be known as the “heads up display” or HUD. In game development, the HUD is one of the most important aspects for conveying information to the player. This information that is being delivered to the player may range from data such as the current health level of the virtual character, the currently equipped weapon, destination markers, enemy information, etc. All this information contributes towards the game feel and immersion of the player, thus always giving them a sense of control when inside the virtual environment and have as much information as possible (Llanos & Jørgensen, 2011). Talking specifically about navigational aids in video games, a key factor for it to be successful is to be able to communicate information regarding where the player currently is, with respect to the environment and location, and communicate future paths that the player can choose to take (Coutrot et. al., 2019).

Traditional games can do this effectively, however when it comes to games in virtual reality, traditional user interface methods are unviable to keep the immersion up due to the way games and simulations are experienced in virtual reality. Immersion and embodiment are two of the key concepts in virtual reality (Wilkinson et al., 2021; Kim & Biocca, 2018; Kanamgotov, 2015) and preservation of it highly depends on the way gameplay elements are implemented in the game or simulation. While designing in virtual reality, the concept of 3D user interface (3DUI) is the framework that is used (Steed et. al., 2021; Bowman et. al., 2001). 3DUI can be explained as an interface developed for use with immersive virtual reality devices where the player is able to manipulate, move, and interact with the elements of the game or simulation as if it was a physical object in the real world. The interesting thing about 3DUI is that all the challenges of the traditional user interfaces does need to be addressed in the world of virtual reality development, along with the interfaces adhering to the basic principles of developing for virtual reality principles of locomotion and selection (LaViola Jr. et. al., 2017), virtual reality immersive program design guidelines (Jerald, 2015), and interaction paradigms (Jacob et. al., 2008).

This leads to an interesting problem to develop, where designers are required to ask the question: how is one supposed to design an integral component, such as a navigational aid, for virtual reality experiences, that still adheres to the rules of the traditional user interface design, but in a way that preserves core virtual reality concepts of immersion and embodiment?

As an example, to understand the general interface design for games in virtual reality (considering only the navigational aid component), a primary survey was done for the top 45 of the bestselling virtual reality video games on the platform Steam (as of October 8, 2021). This was done primarily to understand what classification of navigational aid was being used to assist the player to traverse these worlds in virtual reality. Out of the 45 games, 26 (58%) of the games did not implement any

navigational aid for the player. The challenges for overcoming exploration and traversal were addressed either through level design (such as being contained in a room or a small area). Most of these games did not have a large area to move about in, and only had the player character be active within a small play area. Only 11 (24%) had implemented a navigational aid, that helped players traverse their larger levels. The issue, however, was that very few of these games implemented a 3DUI friendly interface which was interactive and could be said to be a part of the environment itself i.e., be diegetic in nature. Most of the interfaces were conversions of the traditional HUD elements seen in their desktop version counterparts, that appeared as static elements with only passive feedback (spatial images and text panels) being given to the player. This was one of the inspirations behind this paper questioning the fact, that whether navigational aids being implemented as a diegetic element (which will be discussed in the next section) of a virtual reality game can help make it more or equally enjoyable, than a traditional implementation of the aid (such as in the form of a mini map).

2.3. Diegetic Interface Elements in Video Game Navigation Design

The previous section ended with the idea of diegetic elements to be used as user interfaces in virtual reality games and simulations. Thus, these techniques can be labelled to be as diegetic interfaces in video games. This paper will specifically deal with the idea of interactable diegetic navigational aid in virtual reality games and simulations.

The origin of the word diegetic can be traced to “diegesis” from ancient Greek stage plays, which is a type of narrative approach where the elements of the play (or in this case the virtual reality video game) does not hamper the immersion of the actor (here, the player) by breaking the fourth wall (Peacocke et. al., 2015). Iacovides et. al (2015) in their study found that as the skill of the player increases by playing a game, a more cluttered HUD becomes a distraction. For an important aspect of gameplay, such as wayfinding and pathfinding, being able to implement a diegetic navigation aid could prove to be a good approach to designing such less intrusive HUD elements – preserving the concepts of immersion. According to Fagerholt & Lorentzon. (2009), immersion is most appropriate when players “voluntarily adopt the game world as a primary world and reason from the character’s point of view”. According to them, the status of the elements within the game being represented as an in-game element, rather than having a disconnect from the environment, seems to do well in terms of immersion.

In the world of video games, the concept of having diegetic elements was introduced by Galloway (2006). In his conclusion, it was noted that the HUD is a non-diegetic element (after having provided specific definitions of “diegetic” and “non diegetic” in his research). The definition for diegesis in games was “the game’s total world of narrative action” (Galloway, 2006, p. 7). The definition of non-diegetic was stated as “gamic elements that are inside the total gamic apparatus yet outside the portion of the apparatus that constitutes a pretend world of character and story” (Galloway, 2006, p. 7-8). Further research was done on specific genres such as first-person-shooter (FPS) games’ user interfaces. Research done by previously mentioned Fagerholt & Lorentzon (2009), it was stated that game coherence is a crucial factor to consider when implementing diegetic gameplay elements. This decision can be made based on two factors accordingly: existence of the element in the fictional world and its relation to the 3D game space. This is evident

in the studies conducted by Fragoso (2014) and Llanos & Jorgensen (2011) where the results stated that the participant's gameplay, in a game with majorly diegetic interfaces, lacked feedback that was meaningful, and the displayed information was vague, thus preferring the HUD user interface more, even though diegetic interfaces were more appealing.



Figure 2: An example of the game Arizona Sunshine where the navigational aid is a diegetic object in the level (road sign that tells the player where to go). This is still non-interactable. (Vertigo Games and Jaywalkers Interactive, 2016)



Figure 3: In Fallout 4 VR, the device that shows the minimap has buttons that could have been used for interactive purposes (such as pinch and rotate the knob with fingers to zoom in) but it fails to do so. (Bethesda Softworks, 2017)



Figure 4: Into the Radius proposes a system wherein VR where the mini map exists as an object and the player can use a pencil to draw on it and make their own routes. This is the closest system of interactive navigational aid in all the 45 VR games. However, this is still not a diegetic aid. (CM Games, 2019)



Figure 5: Surviv3r game, where the interactive navigational aid exists as a diegetic object in the world in the form of a tablet that the player character can hold. (Candymakers, 2021)

2.4. Diegetic Elements in Virtual Reality and Immersion

From the sections above, diegetic user interfaces are a new concept that are being utilized as novel ways for conveying information to players about the game they are in. One key factor for the need of proper wayfinding and navigation aid elements in virtual environments can be attributed to the fact that presence of extra degrees of freedom and narrow field of view can make the user feel disoriented (Sousa Santos et. al., 2009, Bowman et. al., 1999). In lieu of this, there is a need for proper 3DUI elements, specifically targeted towards navigation aiding and wayfinding, which are a necessary aspect of virtual reality games. Numerous studies have pointed out that diegetic interfaces, which is one of the leading ideas for implementing 3DUI, only work when the immersion is preserved and there is coherence of the element with the environment (Fragoso, 2014; Llanos & Jorgensen, 2011; Fagerholt & Lorentzon, 2009).

Salomoni et. al. (2016) conducted a study to understand the impact diegetic and non-diegetic user interface elements have on the levels of immersion and presence on the user. Their method involved doing the research with shell interfaces (non-diegetic) and global control interfaces (diegetic). The positive side of this research was that this was one of the few studies out there that tried to take a deeper dive into comparing diegetic and non-diegetic interfaces in virtual reality. Their results indicated that in terms of presence, usability, and motion sickness, the diegetic interfaces performed better, and were more appreciated. The diegetic interface was considered more immersive than the non-diegetic one. A flaw of the study could be that the setup used to test the scenario was not on a proper virtual reality system; rather a system comprising multiple different sub-systems. Thus, one of the impacts that could have been on immersion due to unfamiliar control schemes (Llorach et. al., 2014; Moss & Muth, 2011).

Steed et al. (2021) in their study reported on the various 3D user interface (3DUI) techniques that are novel and have either been implemented or are being considered for design and research in this field. They look over the design principles and directions for 3DUI research and a range of factors that need to be considered for 3D user interfaces in virtual reality, be it diegetic or not, to preserve the base concepts of immersion, embodiment, and perception. Various elements of virtual reality game design were discussed such as:

1. Locomotion: joystick walk technique
2. Hand controls and menus: in-hand menus (Lindeman et. al., 1999)
3. Manipulation: snap drop zone (Mapes & Moshell, 1995)
4. Selection: intention-preservation selection (Argelagulet & Andujar, 2009)
5. Framing: diegetic interfaces (Lindeman et. al., 1999)
6. Physics-based interaction: (Steed et. al., 2021)

In the context of this paper, we are considering these techniques' contribution to diegetic interfaces. Steed et. al (2021) give a detailed report of how current virtual reality video games have implemented diegetic elements in them to execute certain features (as an example they cite the game Job Simulator, where instead of pressing a button to exit the game, instead of selecting a button on a traditional menu, the player must eat an "Exit Burrito" to leave the game. They state that most of the academic research is focused on task efficiency rather than reliability, accessibility,

and enjoyment, which is the major focus of consumer games. This paper is a good outlook for design guidelines for developing experiences for virtual-reality-based games and experiences.

The brief survey stated earlier (of the top selling virtual reality games on Steam) has only 2 games with a system for an interactable diegetic interface, that acts as a navigational aid. From the above-mentioned studies, most of the established work focusses on the HUD user interface as a whole unit, rather than considering the individual elements that comprise the HUD and make up for a system that provides the most information to the player. The test cases are also based on traditional video games (i.e., games that are supposed to be experienced through flat display devices) and very few studies have addressed this issue in the field of virtual reality and virtual reality games. Furthermore, the research work also only addresses qualitative assessments mostly, as well as on the immersive of a diegetic element. This paper attempts to focus more on a specific type of HUD element (the navigational aid component) and attempts to discuss whether they can have a pivotal role in affecting the enjoyment of a player in a virtual-reality scenario.

2.5. Presence, Immersion, Enjoyment, and Cybersickness

From the previous sections, much of the current research with diegetic elements in virtual reality (and for that case, virtual reality as a whole) involves having the concept of “immersion” as the dependent variable. In the world of video games, intrinsic motivation is also an important aspect that needs to be considered. From previous studies (Lombard & Ditton, 1997; Wirth et. al., 2007), it has already been established that intrinsic motivation can have a direct impact on the level of presence, and in turn engagement and enjoyment a player is having, while experiencing video games. Studies such as Lee & Kim (2021) and Papagiannidis et. al. (2017) shows that the more interactive and engaging the user is, the more they enjoy the experience in virtual reality. Other studies put forth the relationship between immersion and enjoyment (Schmierbach et. al., 2014; Sylaiou et. al, 2010; Weibel et. al., 2008) thus creating a link between the two and existence of a dependence of each other between these two core virtual reality concepts, thus indicating to the fact that there also needs to be research done in the realm of whether the proposed techniques of diegetic elements, which are known to be immersive, can impart the same level of enjoyment and engagement to the players, when compared to their traditional counterparts. Based on another concept called the “Flow Theory” (Sherry, 2004; Csikszentmihalyi, 2000), it is known that user experience, enjoyment, and engagement can be considered as all contributors to a satisfying experience in virtual reality. Papagiannidis et. al. (2017) and Weibel & Wissmath (2012) in their studies also state the concept of “telepresence” or the feeling of being immersed (in a virtual environment). Thus here, we find a link between telepresence (i.e., the concept of immersion at its core) and enjoyment (through a simulated experience).

Virtual reality began to experience a new dimension and growth as early as 2016, when it was getting attention. Therefore, designers and developers began to research to understand more about what the psychological and physiological effects of could be using virtual reality. Hence Shafer et al., (2019), dived their research into understanding why users react and their engagement with Virtual reality games.

One of the focal points noted by Shafer et al. (2019) and Weibel & Wissmath (2012) is that spatial presence is a valuable factor when it comes to enjoyment of video games. Spatial presence as defined is the feeling that the user is part of the environment built in virtual world. Spatial presence is embodied by realism, which is another factor that makes users enjoy virtual reality games, realism in the context of virtual reality is when the environment is similar in look and feel to what the user has envisioned in his mind. In correlation to this project, we also considered the spatial presence in making it as real as possible for the users and avoid having unwanted elements in the scene causing distractions to the users. Another factor that enforces enjoyment in virtual reality game is valuable at which the user can take control of the game, by mastering his environment and knowing what to do at a particular time. Mastering of the environment comes with user been able to navigate through the spatial presence and which is due to fulfilling two conditions of interactivity which are effectance and control. Hypothetical model has been adopted in most motion-based game to test and validate how the users enjoy the experience, it has been proven to function properly in console games and mobile devices. Enjoyment in VR can be attributed to the level of engagement that the designer/developer have created into the game which is attributed to motivational displayed intended by the interaction (L. Caroux et al., 2015).

Although cybersickness is mentioned as a factor that reduces enjoyment while playing a motion VR game, some of the symptoms of cybersickness include headaches, eyestrain, nausea and in some cases the user ends up throwing up (Abarca et. al., 2020). Cybersickness is synonymous with motion sickness suffered by individuals when sitting in a moving vehicle. In this research there will be little or effects of cybersickness on the users because the movement in this research is paced by the user's itself and when they feel they are moving too fast they can slow down themselves, in some cases it is due to the system failures having some glitches and sending erratic signals during usage based on past research experiences. Some conditions in which the players are positioned trigger cybersickness (Shafer et al., 2019) it is advised that the position in which the user is going to be in virtual reality, should be in the same position in the real world to avoid confliction actions.

3. Methodology

Based on the above screenshots and discussion, the idea that this paper attempts to explore is the implementation of a diegetic navigational aid system, instead of a persistent spatial user interface system, where the user must interact with to obtain navigational information. The final goal being to understand whether players will get more enjoyment out of treating this system as a part of the overall environment, rather than a player-confined system (such as a spatial user interface). In this study, the idea is to explore how interactive diegetic navigation creates an enjoyable game experience for the user compared to non-diegetic maps.

3.1 Study design

For the purpose of the study, a simulation was created to mimic the navigation through a cityscape. The simulation consists of two modes: traditional wrist-mounted minimap method (as explained in Section 2.4) and interactive diegetic console mode. In both modes the following aspects remain the same:

1. The city's roads were constructed to be in a grid-like fashion, with certain points where the roads split up (4 T-junctions in total) and the player must make a decision
2. Each mode has two possible start locations on one end of the map
3. Each mode has two possible end locations, represented as subway stations with specific names (Blue Line and Yellow Line)
4. The method for movement around the roads is point to point teleportation (Section 2.4). There is an added option to move only with joystick (if needed).

At the start of the simulation, the player is assigned a specific start and end point, to develop a “travel from A-to-B” narrative, under a specific time. There are differences in the two modes however:

1. Wrist mounted minimap mode: In this mode, there is a spatial navigation aid, in the form of a minimap, on the virtual wrist of the player. At all times, the player can see their current position, on a top-down projection of the map. The player then must navigate the roads to reach the correct subway to exit the simulation. The minimap only indicates the current position of the player, hence preserving telepresence, and communicating to the player at all times their location with respect to the environment and guiding the player to the destination – thus fulfilling the core task of a navigational aid.
2. Interactive Diegetic Console mode: In this mode, the spatial wrist mounted minimap is replaced. In its place, there exist gates in the world that the player must unlock using a keycard they are equipped with. Based on which subway station they must go to for the simulation to end, the player is equipped with a colored swipe card/security access card. Once they choose a path at a T-junction, they face a closed gate, and must swipe their access card on the console. If they are on the correct path, i.e. say they have to go to End 1 (which is Blue Subway Line) and they have a blue access card, and the gate is on the correct path to End 1, then the player is said to be on the correct choice path. When the correct card is swiped, the console display turns green, and the gate opens up, allowing the player to proceed. If they are on the incorrect path, the console display turns red, and the player must retrace their steps to the last T-junction and choose the other path. In this way, the player must actively interact with their environment to know about telepresence, and understand their position with respect to the environment. The environment will still guide them to the correct end (because even if they are wrong, they will get the correct path after every interaction), thus preserving the core concept of a navigational aid – guiding the player to the destination.

During the simulation, a timer will be present to make sure that the players know that they are under a time crunch to reach the end. That serves as the motivation for the player to concentrate on the task at hand and keeps the player engaged (Shafer et al., 2019; Caroux et al., 2015; Sylaiou et al, 2010), and thus preserve the concept of flow and presence in a video game.

In total, each mode contains four levels for a path configuration (2 starts x 2 ends), thus the total number of treatments available would be 8 (2 factors x 4 levels), out of which each player will go through one condition from each mode (in total 2 treatments, one after another), twice, thus making

it feasible to use an A-B Statistical Testing methodology (Young, 2014). The participants will undergo in the sequence order determined by Graeco Latin Square Design (Sapam & Sinha, 2020). Here the hypothesis would be:

H0 (null): There is no difference in enjoyment levels of the participants, between the two modes.

H1 (alternative): There is a significant difference in enjoyment levels of participants, between the two modes.

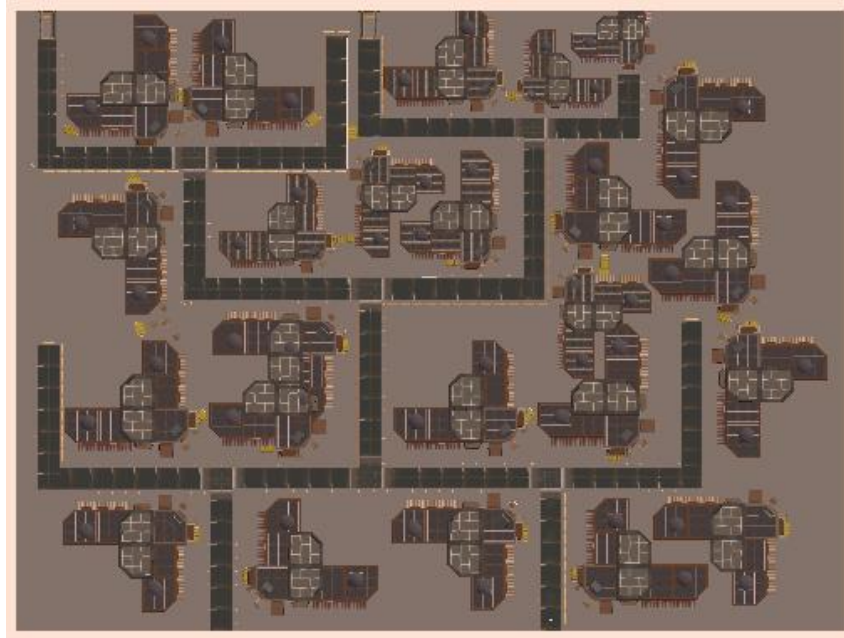


Figure 6: Top-down view of the map. The minimap is also of the same view.



Figure 7: Game view.

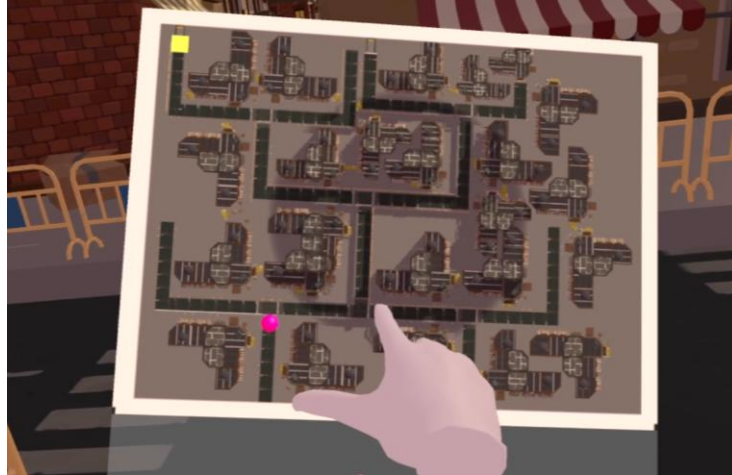


Figure 8: Minimap mode in Virtual Reality.



Figure 9: Diegetic mode. Left shows if correct key is inserted. Right shows if incorrect key is inserted.



Figure 10: Diegetic mode in Virtual Reality. Blue Key in hand indicates the end destination is the Blue Line Subway.

For testing methodology, Table 1 and Table 2 display the Latin Square Design configurations for each mode (M = Minimap mode and D = Diegetic Mode).

M_S1_E1	M_S1_E2	M_S2_E2	M_S2_E1
M_S1_E2	M_S2_E1	M_S1_E1	M_S2_E2
M_S2_E1	M_S2_E2	M_S1_E2	M_S1_E1
M_S2_E2	M_S1_E1	M_S2_E1	M_S1_E2

Table 1: Minimap mode; 1st row shows the first factor with the 4 levels of combination. The rest of the table was achieved through normal Latin Square Design. (M = Minimap Mode, S1 = Start Position 1, S2 = Start Position 2, E1 = End Position 1, E2 = End Position 2)

D_S1_E1	D_S1_E2	D_S2_E2	D_S2_E1
D_S1_E2	D_S2_E1	D_S1_E1	D_S2_E2
D_S2_E1	D_S2_E2	D_S1_E2	D_S1_E1
D_S2_E2	D_S1_E1	D_S2_E1	D_S1_E2

Table 2: Diegetic Mode mode; 1st row shows the second factor with the 4 levels of combination. The rest of the table was achieved through normal Latin Square Design. (D = Diegetic Mode, S1 = Start Position 1, S2 = Start Position 2, E1 = End Position 1, E2 = End Position 2)

Once the two tables were developed, each was treated as a variable (M = {A, B, ...} and D = {1, 2, ...}) and Graeco-Latin Square Design (Sapam & Sinha, 2020) was applied to get the final test ordering for participants.

A1	B2	C3	D4
D2	C1	B4	A3
B3	A4	D1	C2
C4	D3	A2	B1

Figure 11: General configuration layout for Graeco-Latin Square Design.

M_S1_E1	M_S1_E2	M_S2_E2	M_S2_E1		D_S1_E1	D_S1_E2	D_S2_E2	D_S2_E1
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Participant 1	M_S1_E1	D_S1_E1		D_S1_E2	M_S1_E2
Participant 2	M_S2_E1	D_S2_E1		D_S2_E2	M_S2_E2

Figure 12: Final participant order for the first row (all original 8 configurations).

3.2 Hardware Details

The simulation was created in Unity game engine, and is targeted for use in virtual reality devices, specifically the Oculus Quest 2. The simulation will be run in wired mode, connected to a laptop personal computer. The laptop is an Asus Republic of Gamers Scar III model, with an RTX 2070 graphics card, and 16 GB of RAM. The simulation runs at 60 FPS without any noticeable stutters or lag.

3.3 Questionnaire Design

Different approaches have been used in measuring the player experience from the objectives of the game to in-game behavior depending on the sole aim of the researchers. However, in this study we are aiming to measure 3 factors that can determine the enjoyment of a player in game play.

For Immersion, we have picked selected questions from Player Experience Inventory (PXI), was validated by (Abeelee et al., 2020) and we are making use of 7-Point Likert Scale (Likert 1932, Matell et al., 1972).

- I was no longer aware of my surroundings while I was playing (1 to 7)
- I was immersed on the game (1 to 7)
- I was fully focused on the game. (1 to 7)
- The game gave clear feedback on my progress towards the goals (1 to 7)
- I felt a sense of freedom about how I wanted to play this game (1 to 7)

For Enjoyment related questions, we chose questions from the Intrinsic Motivation Inventory (IMI) for its contribution to understanding innate psychological needs—competence, autonomy, and relatedness. Which was validated by (Ryan et al., 2000) and we are making use of 7-Point Likert Scale (Likert 1932, Matell et al., 1972).;

- I enjoyed doing this activity very much (1 to 7)
- This activity was fun to do. (1 to 7)
- I thought this was a boring activity. (1 to 7)
- This activity did not hold my attention at all. (1 to 7)
- I would describe this activity as very interesting. (1 to 7)
- I thought this activity was quite enjoyable. (1 to 7)
- While I was doing this activity, I was thinking about how much I enjoyed it. (1 to 7)

For Control related questions, we picked questions from The Player Experience of Need Satisfaction (PENS) because it has been used to explore different aspect of game settings within recreational game and it has been validated by (Johnson et al., 2018) and we are making use of 7-Point Likert Scale (Likert 1932, Matell et al., 1972).;

- Learning the game controls was easy. (1 to 7)
- The game controls are intuitive. (1 to 7)
- When I wanted to do something in the game, it was easy to remember the corresponding control. (1 to 7)

REFERENCES:

- Abarca, V. M., Palos-Sanchez, P. R., & Rus-Arias, E. (2020). Working in Virtual Teams: A Systematic Literature Review and a bibliometric analysis. *IEEE Access*, 8, 168923-168940. doi:10.1109/access.2020.3023546
- Abeeel, V. V., Spiel, K., Nacke, L., Johnson, D., & Gerling, K. (2020). Development and validation of the Player Experience Inventory: A scale to measure player experiences at the level of functional and psychosocial consequences. *International Journal of Human-Computer Studies*, 135, 102370. doi:10.1016/j.ijhcs.2019.102370
- Argelaguet, F., & Andujar, C. (2009). Efficient 3D pointing selection in cluttered virtual environments. *IEEE Computer Graphics and Applications*, 29(6), 34-43. doi:10.1109/mcg.2009.81
- Bowman, D. A., Davis, E. T., Hodges, L. F., & Badre, A. N. (1999). Maintaining spatial orientation during travel in an immersive virtual environment. *Presence: Teleoperators and Virtual Environments*, 8(6), 618-631. doi:10.1162/105474699566521
- Bowman, D. A., Kruijff, E., LaViola, J. J., & Poupyrev, I. (2001). An introduction to 3-D user interface design. *Presence: Teleoperators and Virtual Environments*, 10(1), 96-108. doi:10.1162/105474601750182342
- Caroux, L., Isbister, K., Le Bigot, L., & Vibert, N. (2015). Player–video game interaction: A systematic review of current concepts. *Computers in Human Behavior*, 48, 366-381. doi:10.1016/j.chb.2015.01.066
- Chen, J., Pyla, P., & Bowman, D. (2004). Testbed evaluation of navigation and text display techniques in an information-rich virtual environment. *IEEE Virtual Reality 2004*, 181-289. doi:10.1109/vr.2004.1310072
- Coutrot, A., Schmidt, S., Coutrot, L., Pittman, J., Hong, L., Wiener, J. M., . . . Spiers, H. J. (2019). Virtual navigation tested on a mobile app is predictive of real-world wayfinding navigation performance. *PLOS ONE*, 14(3), e0213272. doi:10.1371/journal.pone.0213272
- Csikszentmihalyi, M. (2000). *Beyond boredom and anxiety*. San Francisco, CA: Jossey-Bass.
- Dautovic, G. (2021, June 21). Top Video Game Industry Statistics (2021). Retrieved October 01, 2021, from <https://fortunly.com/statistics/video-game-industry-statistics/>
- Evans, A., & Blat, J. (2014). Simulator sickness and presence using hmds. *Proceedings of the 20th ACM Symposium on Virtual Reality Software and Technology - VRST '14*, 137-140. doi:10.1145/2671015.2671120
- Fagerholt, E., & Lorentzon, M. (2009). *Beyond the HUD-user interfaces for increased player immersion in FPS games* (Master's thesis).

- Fragoso, S. (2014). Interface Design Strategies and disruptions of gameplay: Notes from a qualitative study with first-person gamers. *Human-Computer Interaction. Applications and Services*, 593-603. doi:10.1007/978-3-319-07227-2_56
- Galloway, A. R. (2010). *Gaming: Essays on algorithmic culture* (Vol. 18). Minneapolis, MN: University of Minnesota Press.
- Iacovides, I., Cox, A., Kennedy, R., Cairns, P., & Jennett, C. (2015). Removing the HUD. *Proceedings of the 2015 Annual Symposium on Computer-Human Interaction in Play*. doi:10.1145/2793107.2793120
- Jacob, R. J., Girouard, A., Hirshfield, L. M., Horn, M. S., Shaer, O., Solovey, E. T., & Zigelbaum, J. (2008). Reality-based interaction: A framework for post-WIMP interfaces. *In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '08)*, 201-210. doi:10.1145/1357054.1357089
- Jerald, J. (2016). *The VR book: Human-centered design for virtual reality*. New York, NY: Association for Computing Machinery.
- Johnson, D., Gardner, M. J., & Perry, R. (2018). Validation of two game experience scales: The player experience of need satisfaction (pens) and Game Experience Questionnaire (Geq). *International Journal of Human-Computer Studies*, 118, 38-46. doi:10.1016/j.ijhcs.2018.05.003
- Kanamgotov, A. (2015). *A framework for immersion in virtual reality* (Doctoral dissertation) (pp. 1-210). Luton, Bedfordshire: University of Bedfordshire.
- Kim, G., & Biocca, F. (2018). Immersion in virtual reality can increase exercise motivation and physical performance. *Virtual, Augmented and Mixed Reality: Applications in Health, Cultural Heritage, and Industry*, 94-102. doi:10.1007/978-3-319-91584-5_8
- Laviola, J. J. (2017). *3D user interfaces: Theory and practice* (2nd ed.). Boston, MA: Addison-Wesley Professional.
- Lee, W., & Kim, Y. H. (2021). Does VR tourism enhance users' experience? *Sustainability*, 13(2), 806-821. doi:10.3390/su13020806
- Likert, R. (1932). A technique for the measurement of attitudes. *Archives of Psychology*, 22(140), 55. doi:https://psycnet.apa.org/record/1933-01885-001
- Lindeman, R., Sibert, J., & Hahn, J. (1999). Hand-held windows: Towards effective 2D interaction in immersive virtual environments. *Proceedings IEEE Virtual Reality (Cat. No. 99CB36316)*, 205-212. doi:10.1109/vr.1999.756952
- Llanos, Stein C., and Kristine Jørgensen. "Do players prefer integrated user interfaces? A qualitative study of game UI design issues." *Proceedings of DiGRA 2011 Conference: Think Design Play*. 2011.

- Lombard, M., & Ditton, T. (2006). At the heart of it all: The concept of presence. *Journal of Computer-Mediated Communication*, 3(2). doi:10.1111/j.1083-6101.1997.tb00072.x
- Matell, M. S., & Jacoby, J. (1972). Is there an optimal number of alternatives for likert-scale items? effects of testing time and scale properties. *Journal of Applied Psychology*, 56(6), 506-509. doi:10.1037/h0033601
- Moghadam, K., Banigan, C., & Ragan, E. D. (2020). Scene transitions and teleportation in virtual reality and the implications for spatial awareness and sickness. *IEEE Transactions on Visualization and Computer Graphics*, 26(6), 2273-2287. doi:10.1109/tvcg.2018.2884468
- Moss, J. D., & Muth, E. R. (2011). Characteristics of head-mounted displays and their effects on simulator sickness. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, 53(3), 308-319. doi:10.1177/0018720811405196
- Papagiannidis, S., Pantano, E., See-To, E. W., Dennis, C., & Bourlakis, M. (2017). To immerse or not? experimenting with two virtual retail environments. *Information Technology & People*, 30(1), 163-188. doi:10.1108/itp-03-2015-0069
- Pausch, R., Burnette, T., Brockway, D., & Weiblen, M. E. (1995). Navigation and locomotion in virtual worlds via Flight into hand-held miniatures. *Proceedings of the 22nd Annual Conference on Computer Graphics and Interactive Techniques - SIGGRAPH '95*, 399-400. doi:10.1145/218380.218495
- Peacocke, M., Teather, R. J., Carette, J., & MacKenzie, I. S. (2015). Evaluating the effectiveness of huds and diegetic ammo displays in first-person shooter games. *2015 IEEE Games Entertainment Media Conference (GEM)*, 138-145. doi:10.1109/gem.2015.7377211
- Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist*, 55(1), 68-78. doi:10.1037/0003-066x.55.1.68
- Salomoni, P., Prandi, C., Rocchetti, M., Casanova, L., Marchetti, L., & Marfia, G. (2017). Diegetic user interfaces for virtual environments with hmds: A user experience study with oculus rift. *Journal on Multimodal User Interfaces*, 11(2), 173-184. doi:10.1007/s12193-016-0236-5
- Sapam, S., & Sinha, B. K. (2020). Graeco Latin square designs with neighbor effects. *Journal of Statistical Theory and Practice*, 15(1). doi:10.1007/s42519-020-00142-3
- Schmierbach, M., Chung, M., Wu, M., & Kim, K. (2014). No one likes to lose. *Journal of Media Psychology*, 26(3), 105-110. doi:10.1027/1864-1105/a000120
- Shafer, D. M., Carbonara, C. P., & Korpi, M. F. (2019). Factors affecting enjoyment of virtual reality games: A comparison involving consumer-grade virtual reality technology. *Games for Health Journal*, 8(1), 15-23. doi:10.1089/g4h.2017.0190

- Sherry, J. L. (2004). Flow and media enjoyment. *Communication Theory*, 14(4), 328-347. doi:10.1111/j.1468-2885.2004.tb00318.x
- Snider, M. (2021, July 13). Two-thirds of Americans, 227 million, play video games. for many games were an escape, stress relief in pandemic. Retrieved October 01, 2021, from <https://www.usatoday.com/story/tech/gaming/2021/07/13/video-games-2021-covid-19-pandemic/7938713002/>
- Solheim, J. (2021). Open World Navigation and the Effects of Diegetic Elements in Video Games (*Master's thesis*, NTNU).
- Sousa Santos, B., Dias, P., Pimentel, A., Baggerman, J., Ferreira, C., Silva, S., & Madeira, J. (2008). Head-mounted display versus desktop for 3D navigation in virtual reality: A user study. *Multimedia Tools and Applications*, 41(1), 161-181. doi:10.1007/s11042-008-0223-2
- Steed, A., Takala, T. M., Archer, D., Lages, W., & Lindeman, R. W. (2021). Directions for 3D user interface research from consumer VR games. *IEEE Transactions on Visualization and Computer Graphics*, 27(11), 4171-4182. doi:10.1109/tvcg.2021.3106431
- Sylaiou, S., Mania, K., Karoulis, A., & White, M. (2010). Exploring the relationship between presence and enjoyment in a virtual museum. *International Journal of Human-Computer Studies*, 68(5), 243-253. doi:10.1016/j.ijhcs.2009.11.002
- Weibel, D., & Wissmath, B. (2011). Immersion in computer games: The role of spatial presence and flow. *International Journal of Computer Games Technology*, 2011, 1-14. doi:10.1155/2011/282345
- Weibel, D., Wissmath, B., Habegger, S., Steiner, Y., & Groner, R. (2008). Playing online games against computer- vs. human-controlled opponents: Effects on presence, flow, and enjoyment. *Computers in Human Behavior*, 24(5), 2274-2291. doi:10.1016/j.chb.2007.11.002
- Wilkinson, M., Brantley, S., & Feng, J. (2021). A mini review of presence and immersion in virtual reality. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, 65(1), 1099-1103. doi:10.1177/1071181321651148
- Williams, C. (2021, February 24). Nearly half of the global population now plays video games, report finds. Retrieved December 04, 2021, from <https://www.givemesport.com/1653435-gaming-study-reveals-how-much-of-the-global-population-now-plays-video-games>
- Wirth, W., Hartmann, T., Böcking, S., Vorderer, P., Klimmt, C., Schramm, H., . . . Jäncke, P. (2007). A process model of the formation of spatial presence experiences. *Media Psychology*, 9(3), 493-525. doi:10.1080/15213260701283079
- Young, S. W. (2014). Improving library user experience with a/B testing: Principles and process. *Weave: Journal of Library User Experience*, 1(1). doi:10.3998/weave.12535642.0001.101